TUGAS 3 FI1101 MODUL 03

Nomor 3, 11, 13, 14 dan 15

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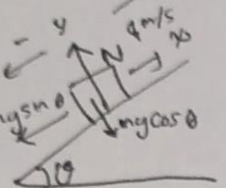
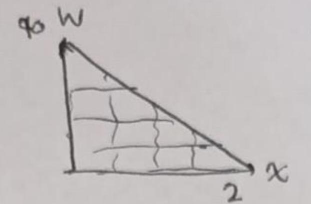
Tanggal : 14 Oktober 2025

Jawaban :

3.



$K_s = \frac{1}{2} m v_0^2$
$\Rightarrow 40 = \frac{1}{2} m \cdot 4^2$
$\Rightarrow m = 5 \text{ kg}$



$dW = F \cdot dx$
$\Rightarrow \frac{dW}{dx} = F$
$\Rightarrow F =$ gradien grafik

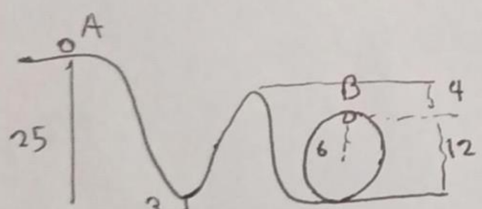
$F = \frac{0-40}{2-0} = -20$ N. Gaya yang bekerja pada sumbu-x hanya $-mg\sin\theta$ sehingga (negatif karena berlawanan arah)

$\Rightarrow F = -mg\sin\theta$
$\Rightarrow -20 = -5 \cdot 9{,}8 \cdot \sin\theta$
$\Rightarrow \sin\theta = \frac{20}{49}$

$\cos\theta = \sqrt{1-\sin^2\theta}$
$= \sqrt{1-\left(\frac{20}{49}\right)^2}$
$= \sqrt{\frac{2001}{2401}} = \frac{1}{49}\sqrt{2001}$

$N - mg\cos\theta = 0$ (Hukum I Newton)
$\Rightarrow N = mg\cos\theta$
$= 5 \cdot 9{,}8 \cdot \frac{1}{49}\sqrt{2001}$
$= \boxed{\sqrt{2001} \text{ N} \approx 44{,}7325 \text{ N}}$

11.

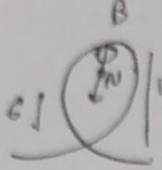


$m = 350$ kg, licin $\Rightarrow E_{MA} = E_{MB} = E_{M\,\text{manapun}}$, $v_A = 0$ m/s
Menggunakan Hukum Kekekalan Mekanik
$E_{M1} = E_{M2} \Rightarrow E_{k1} + E_{p1} = E_{k2} + E_{p2}$

$E_{MA} = mgh_A + \frac{1}{2}mv_A^2$
$= 350 \cdot 9{,}8 \cdot 25$
$= 85750$ J

(a) v_B? $E_{MA} = E_{MB}$
$\Rightarrow 85750 = mgh_B + \frac{1}{2}mv_B^2$
$= 350 \cdot 9{,}8 \cdot 12 + \frac{1}{2} \cdot 350 \cdot v_B^2$
$= 41160 + 175 v_B^2$
$\Rightarrow v_B^2 \cdot 175 = 44590$
$\Rightarrow v_B^2 = \frac{1274}{5} \Rightarrow \boxed{v_B = \frac{7}{5}\sqrt{130} \text{ m/s} \approx 15{,}9624 \text{ m/s}}$

(b)



Setiap gerak melingkar pasti memiliki gaya sentripetal dan sesuai dengan gambar, gaya kontak ini sama persis dengan gaya sentripetal sehingga

$N = \frac{mv^2}{r} \Rightarrow N_B = \frac{m v_B^2}{r}$ dengan $r = 6$ m

$v_B = \frac{7}{5}\sqrt{130}$ m/s

Maka, $N = \frac{350 \cdot \left(\frac{7}{5}\sqrt{130}\right)^2}{6}$

$= \frac{350 \cdot \frac{49}{25} \cdot 130}{6}$

$= \boxed{22.295 \text{ N}}$

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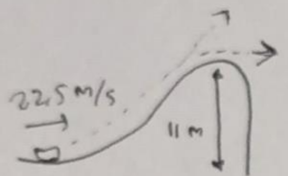
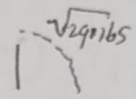
13.



m = 125 kg Pakai Hukum kekekalan energi lian



$t \Rightarrow y = v_y t + \frac{1}{2} g t^2$

$11 = 4{,}9 t^2$

$\Rightarrow t = \sqrt{\frac{11}{4{,}9}} = \frac{1}{7}\sqrt{110}$

$E_{M_0} = E_{M_1}$

$mgh_0 + \frac{1}{2} m v_0^2 = mgh_1 + \frac{1}{2} m v_1^2$

$\Leftrightarrow gh_0 + \frac{1}{2} v_0^2 = gh_1 + \frac{1}{2} v_1^2$

$\Rightarrow \frac{1}{2}(22{,}5)^2 = 9{,}8 \cdot 11 + \frac{1}{2} v_1^2$

$\Rightarrow \frac{1}{2} v_1^2 = 145{,}325$

$\Rightarrow v_1 = \sqrt{290{,}65}$ [ini v_x karena bentuknya $\vec{\int}$]

$\Delta x = v \cdot \Delta t$

$= \sqrt{290{,}65} \cdot \frac{1}{7}\sqrt{110}$

$= \boxed{\frac{1}{7}\sqrt{31971{,}5} \text{ m} \approx 25{,}54368 \text{ m}}$

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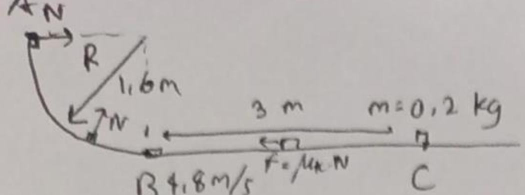
14.



$m = 0,2$ kg

$E_{pB} = E_{pC} = 0$ J

(a) $V_B = 4,8$ m/s, $V_C = 0$

$E_{kb} - E_{kf} = E_{kc}$

$\Rightarrow \frac{1}{2} m v_b^2 - F \cdot \Delta x = 0$

$\Rightarrow F \cdot 3 = \frac{1}{2} \cdot 0,2 \cdot 4,8^2$

$\Rightarrow F = \frac{2,304}{3}$ N

$\Rightarrow f = \frac{2,304}{3}$ N

$N = mg = 0,2 \cdot 9,8$
$= 1,96$ N

$f = \mu_k \cdot N$

$\Rightarrow \mu_k = \frac{f}{N}$

$= \frac{\frac{2,304}{3}}{1,96}$

$= \boxed{\frac{2,304}{5,88} \approx 0,3918}$

2

$E_{MB} - E_{MA}$ = Akhir − awal

(b) $W_f = -(E_{MA} - E_{MB})$

$= -((\frac{1}{2} m v_A^2 + m g h_A) - (\frac{1}{2} m v_B^2 + m g h_B))$

$= -(mgR - \frac{1}{2} m v_B^2)$

$= -0,2 (9,8 \cdot 1,6 - \frac{1}{2} \cdot (4,8)^2)$

$= -0,2 (4,16)$

$= \boxed{-0,832 \text{ J}}$ (atau +0,832 J karena menguangi energi)

$E_{MA} - E_{MB}$

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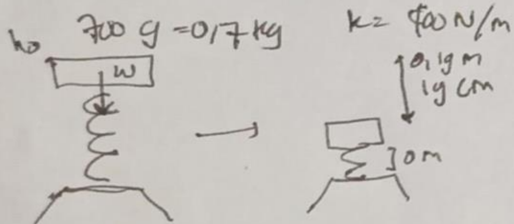
15.



700 g = 0,7 kg

k = 400 N/m

$w = mg$
$= 0,7 \cdot 9,8$
$= 6,86$ N

(a) $W_{kp} = \frac{1}{2} k \Delta x^2$

$= \frac{1}{2} \cdot 400 \cdot (0,19)^2$

$= 200 \; (0,0361)$

$= \boxed{7,22 \text{ J}}$

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(b) Menggunakan Hukum III Newton, didapat $W_{pk} = -W_{kp}$

$= \boxed{-7,22 \text{ J}}$

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(c) $E_{M0} = E_{M1}$

$\Rightarrow mg(h_0 + \Delta x) = \frac{1}{2} k \Delta x^2$

$\Rightarrow 0,7 \cdot 9,8 \; (h_0 + 0,19) = 7,22$

$\Rightarrow h_0 + 0,19 = \frac{7,22}{6,86}$

$\Rightarrow \boxed{h_0 = \frac{7,22}{6,86} - 0,19 \text{ m} \approx 0,8624 \text{ m atau } 86,24 \text{ cm}}$

2 3

(d) Menggunakan Hukum Kekekalan Energi,

$E_{M_0} = E_{M_1}$

$\Rightarrow mg(2h_0 + \Delta x') = \frac{1}{2} k (\Delta x')^2$

$\Rightarrow 2h_0 + \Delta x' = \frac{200 \, \Delta x'^2}{6{,}86}$

$\Rightarrow \frac{200}{6{,}86} \Delta x'^2 - \Delta x' - 2h_0 = 0$

$\Rightarrow \frac{200}{6{,}86} \Delta x'^2 - \Delta x' - 2\left(\frac{7{,}22}{6{,}86} - 0{,}19\right) = 0$

Gunakan rumus ABC dengan $a = \frac{200}{6{,}86}$, $b = -1$, dan $c = \left(0{,}19 - \frac{7{,}22}{6{,}86}\right) \cdot 2$

$$\Delta x' = \frac{1 \pm \sqrt{1 - 4 \cdot \frac{200}{6{,}86} \left(0{,}19 - \frac{7{,}22}{6{,}86}\right) 2}}{2 \cdot \frac{200}{6{,}86}}$$

= (Ambil akar positif $\Rightarrow \pm \to +$)

$$= \frac{1 + \sqrt{1 + \frac{800}{6{,}86} \left(\frac{7{,}22}{6{,}86} - 0{,}19\right) 2}}{\frac{200}{3{,}43}}$$

$\approx \boxed{0{,}260994 \text{ m} = 26{,}0994 \text{ cm}}$